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Student Name: Faheem Dehalvi Student ID: S4197364

Data Preparation

The raw victorian_road_crash_data.csv was loaded using pandas with engine='python' to preserve correct string dtypes across all 150,452 rows and 52 columns. Before any cleaning was applied, every column was data profiled — value counts for all categorical fields and descriptive statistics for all numeric columns so we could understand what steps are to be taken to clean the data. Below each step of data cleaning is mentioned.

Date and Time Formatting

ACCIDENT_DATE was stored as a plain string (e.g. "1/1/2015"), which prevents any year or month-based filtering in pandas. It was converted to a proper datetime object using pd.to_datetime with dayfirst=True to match the Australian day-first convention.

ACCIDENT_TIME was also stored as a string. Profiling identified 27 rows where the hour portion exceeded 23 — values such as "30:00.0" and "55:00.0" are physically impossible on a 24-hour clock and are data entry errors. Then these 27 rows were corrected in-place by replacing the bad strings with the column mode time ("15:00:00").

Typos and Inconsistent Labels

Several categorical columns contained label quality issues discovered through value count profiling. In ACCIDENT_TYPE, one row held "." as a placeholder with no informational value, and the value "collision with some other object" used a lowercase first letter while all other types were title-cased — causing it to register as a separate category in any groupby.

The placeholder was replaced with mode which was 'Collision With Vehicle', .str.title() was applied across the whole column to normalise casing, and the single '.' Was replaced with mode.

In DAY_OF_WEEK, "Thu" appeared once as a short form for "Thursday" and was directly mapped to the full name.

"Weekend" is not a specific day and cannot be used in day-level analysis for graps; it was replaced with the mode of existing Saturday and Sunday rows, which resolved to "Saturday".

In LIGHT_CONDITION, "Days" was a clear typo of "Day" and was corrected directly.

"Unk." appeared 4,335 times as an informal abbreviation with no data dictionary counterpart, and "No" appeared once with no valid category match — both were converted to NaN and filled with the column mode ("Day").

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Impossible Values and Outliers

The SPEED_ZONE column contained values that do not correspond to any real Victorian speed zone.

The data dictionary confirms 110 km/hr as the legal maximum. Profiling found “10000 km/hr” and “1000 km/hr” (physically impossible data entry errors).

“0” (no Victorian zone is zero),

“30km/hr” (missing the space used by all other labels).

A cleaning function was written to extract the numeric portion of each value, cap anything above 110 to 110.

Treat zero as “Not known”, and reformat all entries consistently as “X km/hr”.

The three non-numeric categories (“Not known”, “Other speed limit”, “Camping grounds or off road”) were passed through unchanged.

Missing Values

For categorical and integer-count columns with missing values, mode imputation was applied.

The mode preserves the existing distribution of a column without introducing artificial numeric averages, and dropping entire crash records over a single missing field was rejected because it would silently remove thousands of otherwise complete rows.

One additional issue was found: STAT_DIV_NAME contained one row with lowercase “unknown”, which was treated as equivalent to null and filled alongside the genuine missing values.

Columns filled using mode: ROAD_NAME (218), ROAD_TYPE (2,105), LGA_NAME (53), DTP_REGION (48), DEG_URBAN_NAME (948), RMA (6,732), DIVIDED (6,732), STAT_DIV_NAME (970), SRNS (105,455), and NO_OF_VEHICLES, HEAVYVEHICLE, PASSENGERVEHICLE, MOTORCYCLE, PT_VEHICLE (9 each). SRNS, RMA, and DIVIDED were retained despite high null rates as they don’t contribute to Task 2 analysis.

For the four coordinate columns (LATITUDE, LONGITUDE, VICGRID_X, VICGRID_Y), each with 48 missing values across the same 48 rows, mean imputation was used instead.

Coordinates are continuous measurements — using the mode would pin all missing crashes to one specific road location. The mean places them at the geographic centroid of the full dataset, which is a neutral choice that does not distort spatial analysis.

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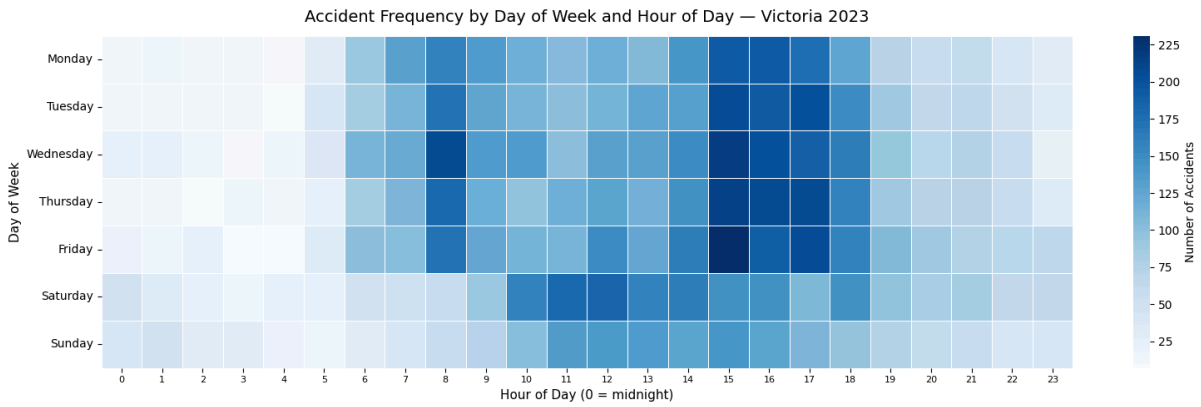
With only 48 of 150,452 rows affected (0.03%), the impact is negligible. The cleaned dataset was saved as cleaned_version.csv.

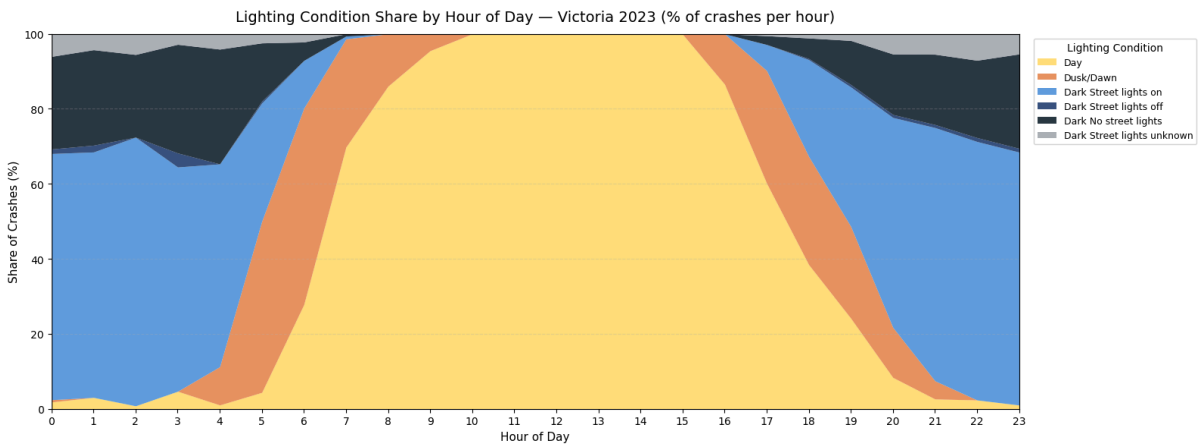
Data Exploration

Task 2.1

The 2023 subset contained 15,532 records. Three separate visualisations were used rather than one combined chart, because a single chart attempting to show day (7 values), hour (24 values), and lighting (6 categories) simultaneously would be unreadable.



A heatmap of Day of Week against Hour of Day was used to show the full 7x24 frequency grid. Weekdays showed two clear peaks: a morning peak around 8am and a stronger afternoon peak between 3pm and 5pm, both consistent with commuter traffic. Saturday evenings (6–9pm) were elevated compared to other weekend hours, while Sunday mornings were the quietest period across the entire week.



A 100% stacked area chart showed how the proportion of lighting conditions shifts across the 24-hour day. This was chosen over a stacked bar because the continuous form better communicates the gradual transition from daylight to dark.

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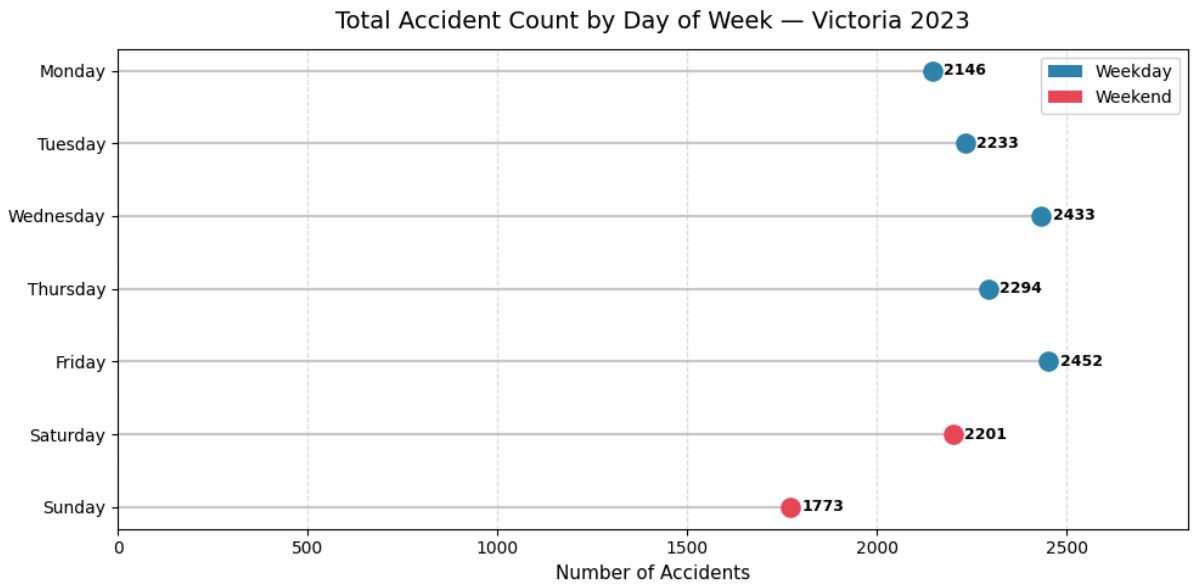
Before 6am and after 7pm, dark conditions (primarily “Dark Street lights on”) dominate. The 6–8am and 5–7pm windows show a clear handoff between “Day” and dark conditions, with “Dusk/Dawn” as a narrow band at the margins.

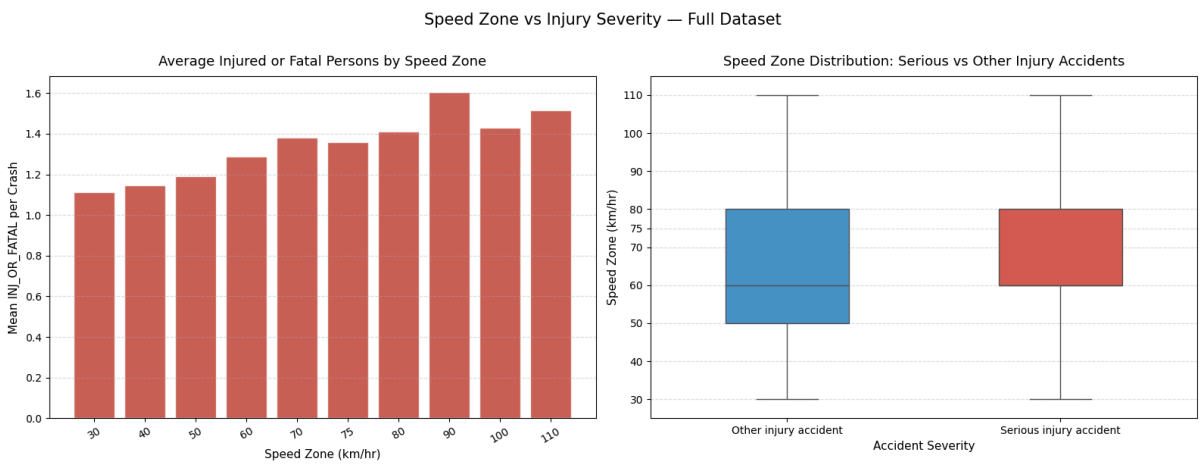


A lollipop chart showed total volume by day — Friday and Wednesday were highest on weekdays; Sunday was the lowest overall.

So final insight, Weekdays show a clear morning peak (~8am) and stronger afternoon peak (3-5pm). Saturday evenings are elevated; Sunday mornings are the quietest period of the week.

Task 2.2

To explore the relationship between speed zones and injury outcomes, a numeric speed zone variable was extracted from the SPEED_ZONE column using regex, enabling quantitative grouping and comparison.



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Two charts were produced side by side. On the left, a bar chart showed the average number of injured or fatal persons per crash at each speed zone.

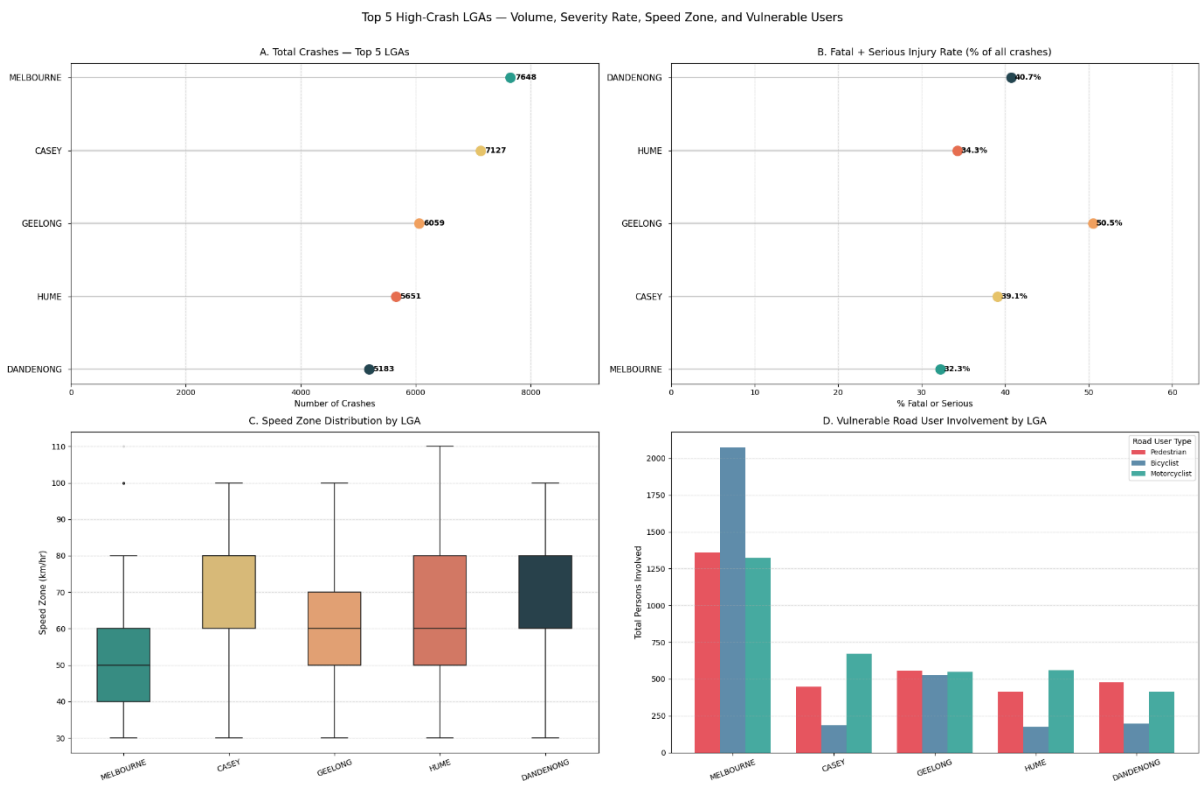
The trend rises clearly with speed — crashes in 100 and 110 km/hr zones produce on average more than twice the injury count of crashes in 50 km/hr zones. This is consistent with the physics of kinetic energy, which scales with the square of velocity, meaning even moderate increases in speed result in significantly greater impact force.

On the right, a boxplot compared the speed zone distribution for "Serious injury accident" versus "Other injury accident". The serious injury group shows a wider spread into higher speed zones, with its median sitting noticeably above that of the other injury group, which concentrates around the lower urban speed zones of 50–60 km/hr.

This confirms that speed zone is a meaningful predictor of crash severity — crashes occurring in high-speed zones are substantially more likely to result in serious injuries regardless of other factors.

Task 2.3

The five LGAs with the highest crash counts were identified as Melbourne, Casey, Geelong, Hume, and Dandenong. A 2x2 grid of visualisations was produced, with a consistent colour palette across all four subplots to make comparisons between LGAs intuitive. The two risk factors examined were severity rate and speed zone distribution, with vulnerable road user involvement included as an additional dimension.



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Plot A used a lollipop chart to show total crash volume per LGA. Melbourne and Casey recorded the highest counts, driven by traffic density rather than inherent road danger. Raw volume alone does not indicate risk, which is why Plot B was necessary.

Plot B showed the fatal and serious injury rate as a percentage of each LGA's total crashes using a dot plot. Geelong stood out with a rate above 50%, despite having fewer total crashes than Melbourne or Casey. Melbourne had the lowest rate of the five, reflecting that the majority of its crashes occur at low urban speeds with less severe outcomes.

Plot C used a boxplot to show the speed zone distribution for each LGA. Geelong and Hume had noticeably higher median speed zones than Melbourne, meaning a greater proportion of their crashes occur on 80–100 km/hr roads. This directly explains the elevated serious injury rates seen in Plot B — higher speeds produce greater kinetic energy at impact.

Plot D used a grouped bar chart to compare pedestrian, bicyclist, and motorcyclist involvement side by side across the five LGAs. Melbourne showed by far the highest pedestrian and cyclist counts, consistent with the density of foot traffic and cycling infrastructure in the CBD. Motorcyclist involvement was more evenly distributed across all five LGAs. Outer LGAs like Casey and Hume had relatively low pedestrian counts, reflecting car-dependent suburban environments where vehicle-to-vehicle collisions dominate.

Taken together, the four plots tell a consistent story: high crash volume in urban LGAs is a product of traffic density and tends to produce lower-severity outcomes, while LGAs with higher-speed road networks carry a disproportionate share of fatal and serious injuries even at lower total crash volumes.

References and Acknowledgement:

1. The dataset was taken directly from canvas and no other references have been quoted in this report

2. Gen AI (ChatGPT) was used to figure out how to loop the data profiling and further also used to phrase my findings in the report and like shape them well and organise it. I used it to learn how some parts could be automated for cleaning like formats and other stuff and yeah, I also asked a lot of doubts to it while I was working on this assignment.